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Visualizing the Dementia Population

Team VizRD:

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DSCI 554: Data Visualization

"Dementia is not an inevitable part of aging — it is a disease, and diseases can be prevented, treated, and someday cured."

— World Health Organization (WHO) public health statement

Understanding the Challenge

- Dementia is a **growing public health crisis** as the U.S. population ages.
- Risk factors and disparities are **complex** and unevenly distributed geographically.

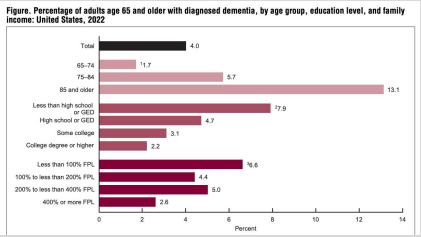
Goal

- **Interactive tool** to map dementia prevalence and risk factors
- **Combines data** on demographics, health, Medicare, and environment
- **Identifies hotspots** where **dementia burden** is greatest
- Supports research, healthcare planning, and policymaking

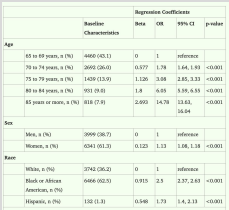
Image Source: Designed by vectorjuice / Freepik (link)

Related Work

Static Tables / Visuals

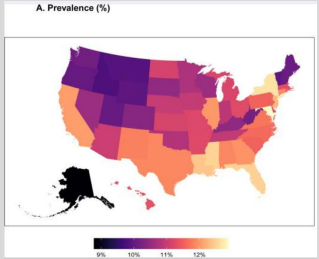


Static bar chart



		Baseline Characteristics	Regression Coefficients: Beta	OR	95% CI	p-value
Age						
	65 to 69 years, n (%)	4460 (45.1)	0	1	reference	
	70 to 74 years, n (%)	2693 (26.0)	0.577	1.78	1.64, 1.93	<0.001
	75 to 79 years, n (%)	1439 (13.9)	1.126	3.08	2.85, 3.33	<0.001
	80 to 84 years, n (%)	931 (9.0)	1.8	6.05	5.59, 6.55	<0.001
	85 years or more, n (%)	818 (7.9)	2.693	14.78	13.63, 16.04	<0.001
Sex						
	Men, n (%)	3999 (38.7)	0	1	reference	
	Women, n (%)	6341 (61.3)	0.123	1.13	1.08, 1.18	<0.001
Race						
	White, n (%)	3742 (36.2)	0	1	reference	
	Black or African American, n (%)	6466 (62.5)	0.915	2.5	2.37, 2.63	<0.001
	Hispanic, n (%)	132 (1.3)	0.548	1.73	1.4, 2.13	<0.001

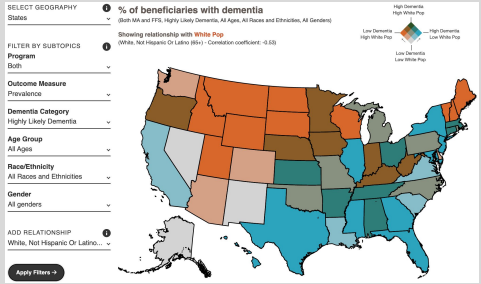
Static Table



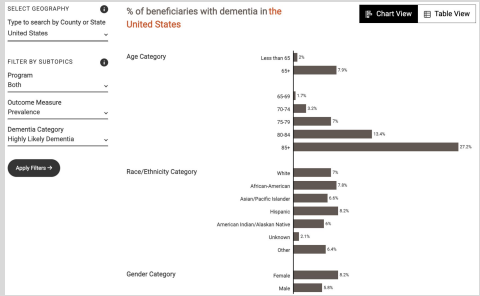
Static choropleth map

Image Source: Zhang et al., 2023, *Frontiers in Neuroscience*

Usability Issues



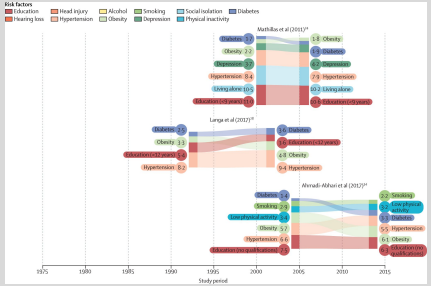
Interactive choropleth map



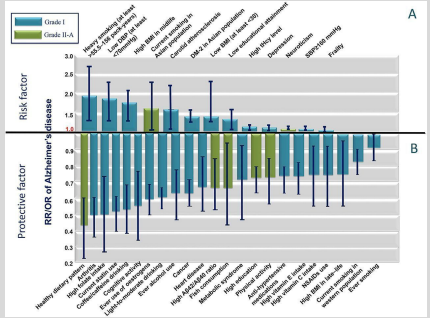
Bar chart

Image Source: NORC at the University of Chicago. (2025). *Interactive Map: Dementia DataHub.*

Limited Interactivity



Bubble chart



Diverging bar chart

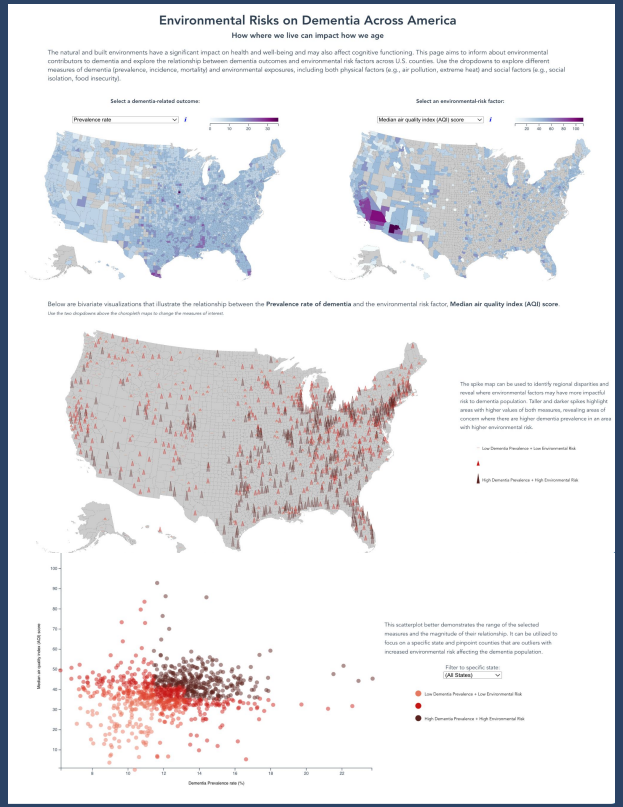
Data

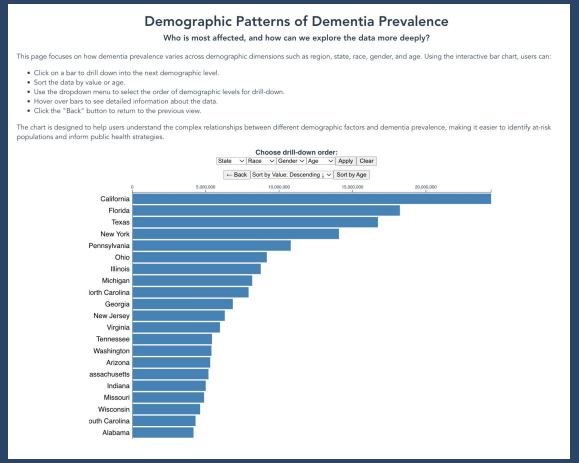
NORC Dementia Datahub	Dementia prevalence for county/state
CDC PLACES Dataset	County-level estimates of chronic conditions and health behaviors
Annual AQI by County	Annual air quality scores for each U.S. county
US County Temperatures & Precipitation	County-level average temperature and precipitation data

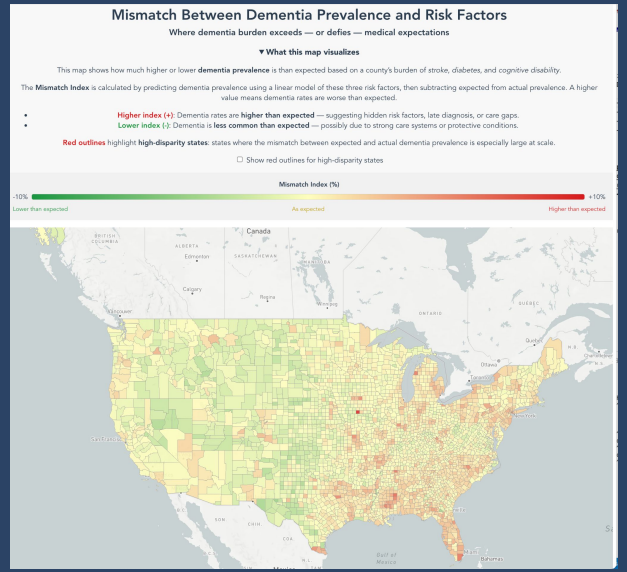
Data

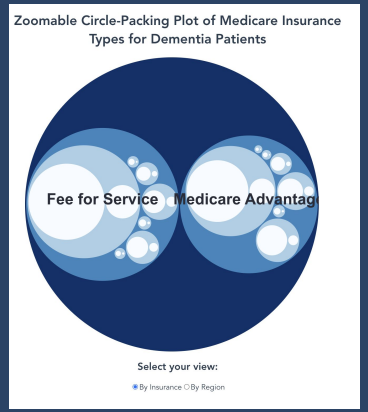
Team VizRD
Overview
Risk Factor Hotspots
Demographic Patterns
Medicare Coverage
Environmental Risks
Visualizing the Dementia Population
Team VizRD: Veasna T. Pusateri, Richard Gallardo, Doris Xia, Damita Gomez
Background & Purpose
Understanding the Challenge
Dementia, including Alzheimer's disease, is a growing public health crisis affecting millions of older adults in the United States. As the population ages, the burden of dementia is expected to rise sharply—impacting families, healthcare systems, and communities. However, the factors that contribute to dementia risk and disparities in care remain complex and unevenly distributed across the country.
Our Solution
This interactive tool was developed to explore the spatial and contextual dimensions of dementia in the U.S. By combining prevalence estimates with data on demographics, health risk factors, Medicare coverage, and physical environmental / socioeconomic risks, the tool identifies where the burden is greatest and what factors may be driving it. Users can interactively explore a variety of measures such as age, race, comorbidities, environmental risks and other factors relate to dementia risk—supporting research, healthcare planning, and informed policymaking.
Sections at a Glance
Data Sources

Dashboard









Demo Link

Individual Contribution

Image Source: Freepik (link)

Limitations & Future Work

Thank you!